2017 F=ma Exam: Problem 20

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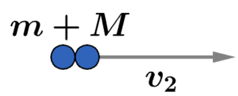
We are given

$$f = \frac{Mv_2}{mv_0}$$

where v_2 is the velocity of M after the collision.



For a perfectly inelastic collision, we have

$$v_2 = \frac{mv_0}{m + M}$$

Then

$$f = \frac{M}{mv_0}\left(\frac{mv_0}{m + M}\right) = \frac{M}{m + M} = \frac{1}{1 + r}$$

where $r = m/M$. To maximize f, we want to minimize r so we take $r \ll 1$. Thus, the answer is $\boxed{\text{A}}$.